

Mayank Chauhan

(Senior Backend and Gen AI Engineer)

PROFESSIONAL SUMMARY

Experienced Backend, Data, and AI Engineer with 4+ years of expertise in designing and building scalable backend systems and data pipelines using Python, Django, and FastAPI. Strong background in ETL data transformation, schema validation, and workflow orchestration for large-scale applications. Skilled in Generative AI and Large Language Models (LLMs), including enterprise-grade implementations using LangChain, LangGraph, and Azure OpenAI. Proficient in Dockerized deployments, CI/CD pipelines, and production-ready AI systems, with a strong focus on performance, reliability, and modern industry best practices.

WORK HISTORY

Chetu India - Software Engineer
Noida
12/2021 - 01/2026

- Developed applications using Generative AI, focusing on large language models to address various business use cases
- Integrated OpenAI API to build chatbots utilizing Retrieval-Augmented Generation (RAG) techniques, optimizing token usage to reduce costs
- Leveraged open-source LLM models for diverse applications, efficiently managing multiple API requests through GPU parallelism and batch processing
- Independently managed projects from inception to deployment on AWS, ensuring seamless implementation and scalability.
- Architected and optimized scalable backend systems using Python, Django, and FastAPI, consolidating distributed services into a maintainable architecture that improved API response times by 40% and supported 1000+ daily transactions.
- Designed and implemented advanced Python-based ETL data transformation frameworks for existing data pipelines, applying schema validation, batch processing, data normalization, and performance optimization for large-scale transactional and analytical workloads.
- Developed Agentic and Agentic RAG applications using single-agent and multi-agent architectures, including AI-driven sales lead automation agents that leveraged enterprise-grade small language models to convert leads into measurable revenue outcomes.
- Built and deployed LLM-powered backend services using LangChain, LangGraph, Azure OpenAI, and Retrieval-Augmented Generation (RAG), enabling context-aware, secure, and production-ready AI integrations.
- Optimized backend performance through database schema redesign, indexing, query tuning, and trigger optimization, improving query efficiency by 35% and reducing infrastructure load.
- Led secure cloud deployments across AWS EC2, AWS ECS, Azure Virtual Machines, bastion servers, VPCs, and VNETs, ensuring 99.9% uptime for Production and UAT environments.
- Implemented CI/CD pipelines using Azure DevOps, Git-based branching strategies, and Dockerized deployments, reducing deployment time from hours to minutes and minimizing merge conflicts by 30%.
- Strengthened application and infrastructure security by implementing JWT authentication, RBAC, WAF, load balancing, network isolation, and regular cyber security assessments.
- Established centralized monitoring, logging, and alerting using AWS CloudWatch and Azure monitoring services, reducing mean time to resolution (MTTR) by 30%, while actively contributing within Agile Scrum teams.

Holitech India Pvt. Ltd. - Data Process Engineer
Noida
03/2020 - 01/2021

- Performed data cleaning, preprocessing, and analysis on NG (non-conforming) product datasets using Python and SQL, identifying root causes, anomalies, and quality trends to support data-driven process improvements.
- Applied statistical analysis, aggregation, and trend analysis on structured production data, contributing to a 10% reduction in defect rates through actionable insights and corrective measures.

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EDUCATION

07/2019
AKTUniversity
Ghaziabad

Bachelor's Degree: Electrical and Electronics Engineer

SKILLS

- Python, JavaScript, SQL, Object-Oriented Programming (OOPs)
- Django, Django REST Framework, FastAPI, RESTful API Development
- Large Language Models (LLMs), Small Language Models (SLMs)
- Generative AI, Agentic AI, Autonomous AI Agents
- Retrieval-Augmented Generation (RAG), Agentic RAG
- LangChain, LangGraph, Azure OpenAI, OpenAI APIs, Transformers
- Prompt Engineering, Conversational AI, Contextual Memory
- ETL Pipelines, Data Engineering, Data Transformation, Data Normalization
- Schema Validation, Batch Processing, Pandas
- PostgreSQL, MySQL, Vector Databases
- AWS (EC2, ECS, CloudWatch), Azure Virtual Machines, VNETs
- Docker, Containerization, Load Balancing, Bastion Servers, VPC
- Azure DevOps CI/CD, Continuous Integration & Deployment
- Git, Version Control, Git-based Workflows
- Search APIs, Google Maps API, LangSmith, Matplotlib
- Linux, Windows Environments

LANGUAGES

- English
- Hindi

Projects

- **Project: Multi-Tenant AI Chatbot for Logistics Platform**

Brief: AI-powered multi-tenant conversational chatbot designed for a logistics platform to automate customer interactions across local, intrastate, and interstate logistics services using contextual LLM-driven workflows.

Technique: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Fine-Tuning, Conversational AI, Contextual Memory, Prompt Engineering, Chat History Management, Geolocation Intelligence.

Tools: Python, Django REST Framework, OpenAI APIs, LangChain, Vector Databases, Google Maps API, AWS (EC2), Generative AI.

Achievements: Successfully deployed a scalable multi-tenant AI chatbot that autonomously captured, qualified, and routed customer requirements, enabling AI-driven lead generation and significantly improving customer engagement efficiency.

- **Project: ETL Data Transformer & Parser for Identity Verification Systems**

Brief: Enterprise-grade Python-based ETL framework for parsing, transforming, merging, and normalizing structured and unstructured data across identity verification pipelines including Facial Recognition, vID, NRD, and Name Change ID systems.

Technique: Advanced ETL Data Transformation, Schema Mapping & Validation, Data Normalization, File Parsing Automation, Reusable Transformation Logic, High-Volume Batch Processing.

Tools: Python (Advanced Concepts), Pandas, Custom ETL Frameworks, Utility Classes, Reusable Transformation Modules, Validation Engines.

Achievements: Delivered a highly reliable ETL solution that improved data accuracy and consistency across multiple identity verification workflows, reduced manual intervention, and accelerated downstream analytics and AI-based identity checks.

- **Project: Agentic AI Platform for B2B Sales Automation**

Brief: Agentic AI-based B2B sales automation platform designed to autonomously identify, enrich, qualify, and convert sales leads using intelligent single-agent and multi-agent workflows with human-in-the-loop feedback.

Technique: Agentic AI Architecture, Multi-Agent Orchestration, Autonomous Decision Making, LLM Pipelines, Vector Search, Prompt Engineering, Lead Enrichment, Sales Intelligence Automation.

Tools: Python, LangChain, LangGraph, Large Language Models (LLMs), Transformers, Vector Databases, Search APIs, LangSmith, Matplotlib, Production Deployment Pipelines.

Achievements: Automated the complete B2B lead generation and qualification lifecycle, significantly reducing manual sales effort while improving lead quality, conversion rates, and revenue outcomes.